# Rubin k8s/batch usage

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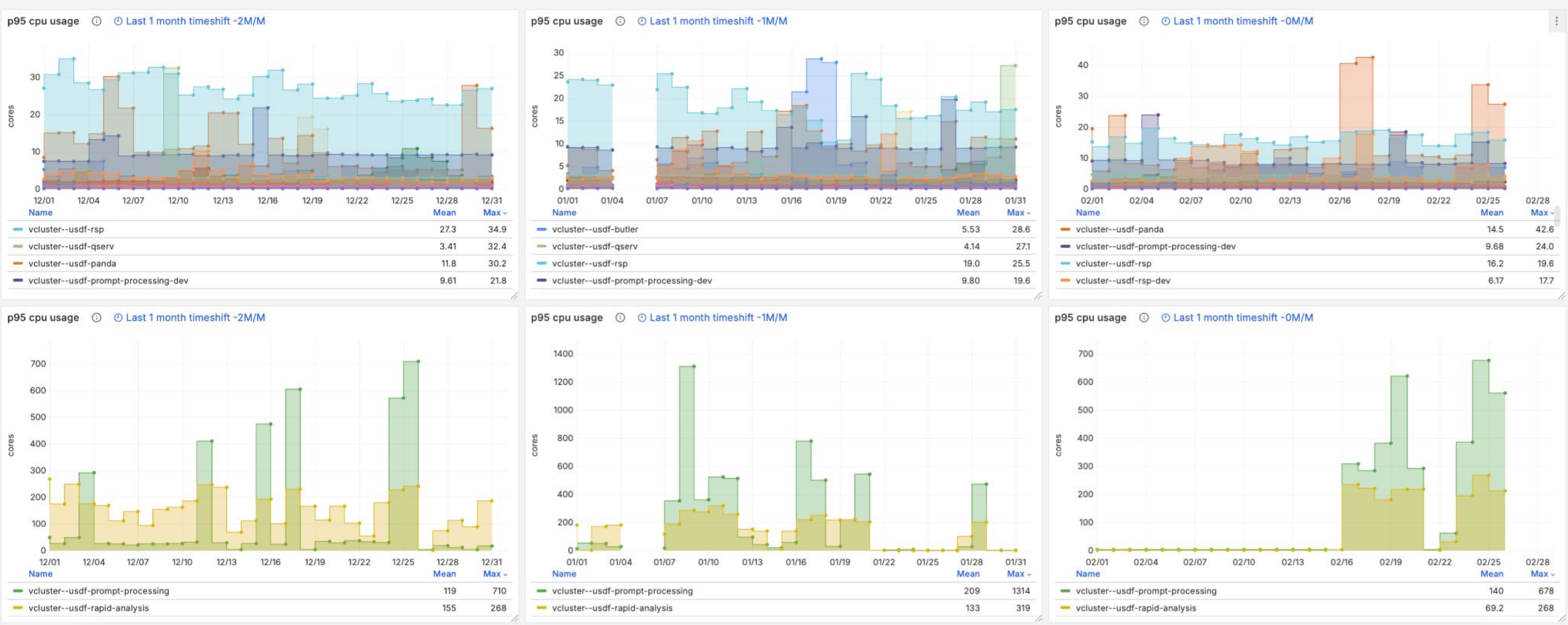
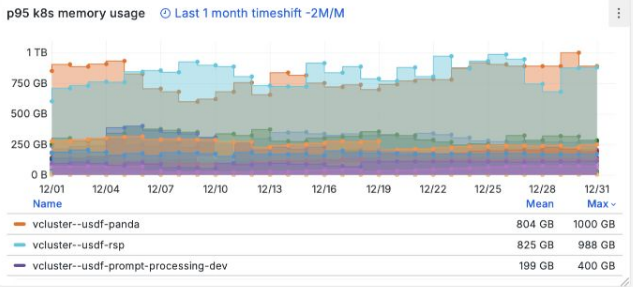
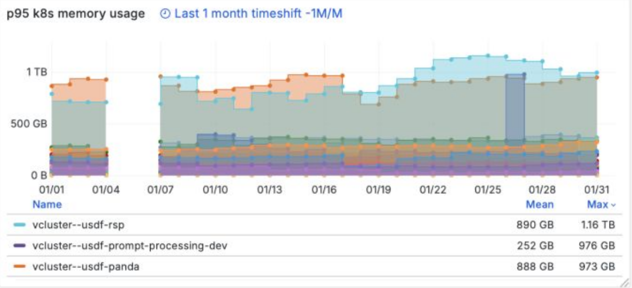
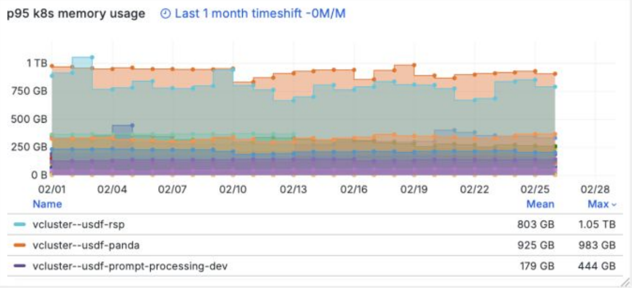
# p95 k8s cpu usage (last 3 months)

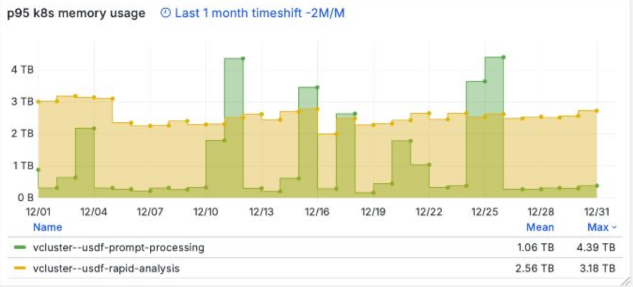


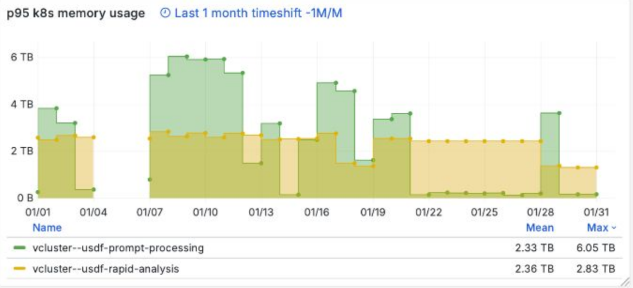
# p95 k8s memory usage (last 3 months) months)

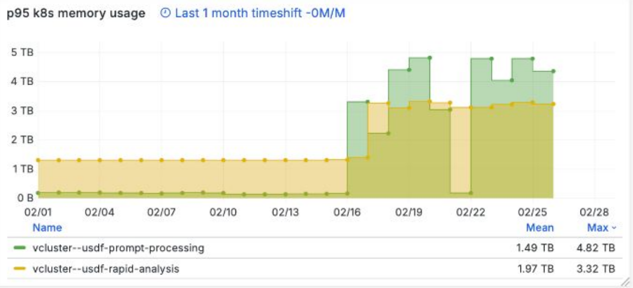


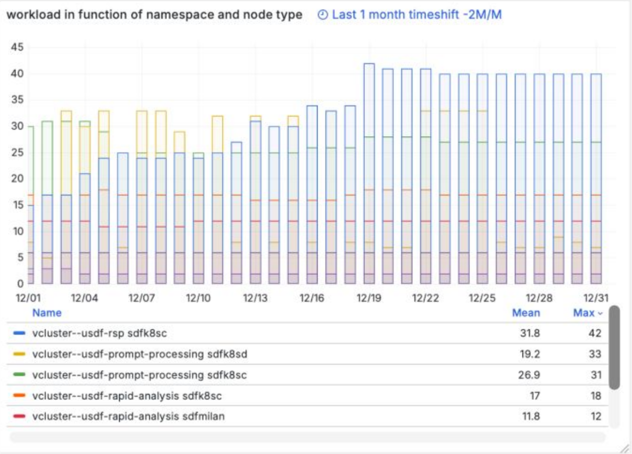



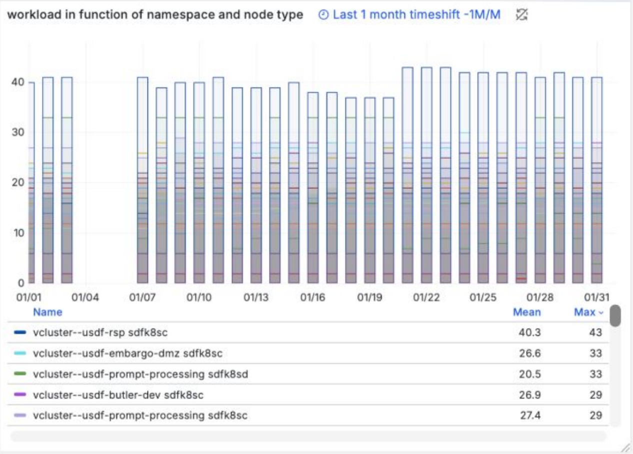



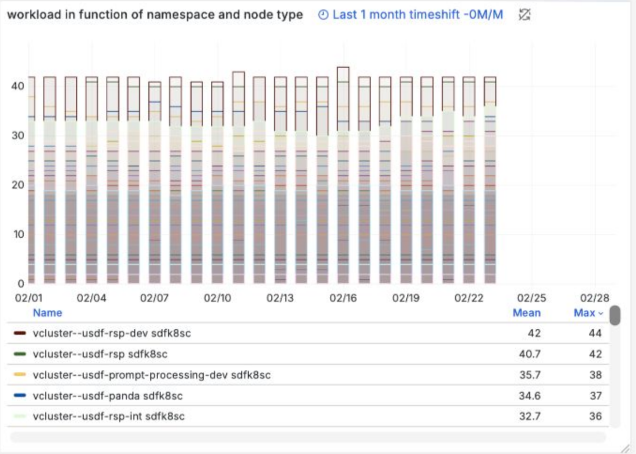



# K8s workload in function of node type



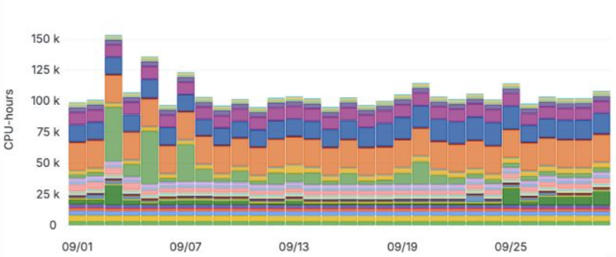



# K8s node specs

| Type | Count | CPU | Mem [GB] |
|---|---|---|---|
| sdfk8sc | 52 | 64 | 256 |
| sdfk8sd | 33 | 96 | 512 |
| sdfk8se | 3 | 32 | 128 |
| sdfk8so | 6 | 128 | 512 |
| sdfmilan* | 12 | 128 | 512 |

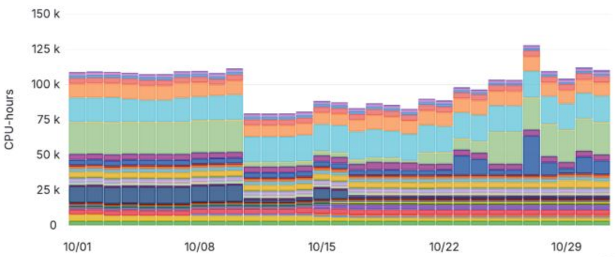
# Kubernetes CPU-Hour usage from kube_pod_limits* (last 6 months)

Container cpu limits ⏲ Last 1 month timeshift -5M/M



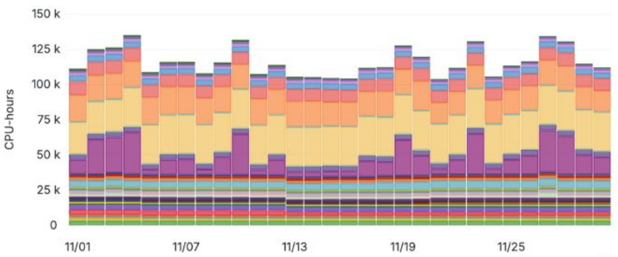

| Name | Mean | Max |
|---|---|---|
| vcluster--usdf-prompt-processing | 9.26 k | 43.8 k |
| vcluster--usdf-rapid-analysis | 22.4 k | 22.5 k |
| vcluster--usdf-rsp | 15.5 k | 18.8 k |
| vcluster--usdf-consdb | 4.53 k | 16.5 k |

Container cpu limits ⏲ Last 1 month timeshift -4M/M



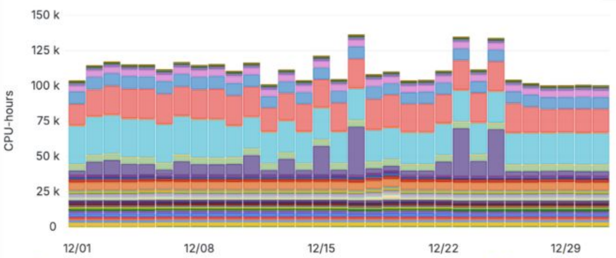

| Name | Mean | Max |
|---|---|---|
| vcluster--usdf-prompt-processing | 5.73 k | 27.9 k |
| vcluster--usdf-rapid-analysis | 14.1 k | 22.4 k |
| vcluster--usdf-rsp | 18.2 k | 20.1 k |
| vcluster--usdf-consdb | 5.26 k | 12.1 k |

Container cpu limits ⏲ Last 1 month timeshift -3M/M



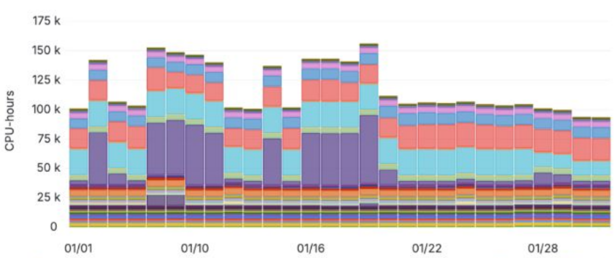

| Name | Mean | Max |
|---|---|---|
| vcluster--usdf-prompt-processing | 13.3 k | 29.5 k |
| vcluster--usdf-rapid-analysis | 26.8 k | 27.4 k |
| vcluster--usdf-rsp | 17.7 k | 20.8 k |
| vcluster--usdf-rsp-dev | 8.88 k | 9.56 k |

Container cpu limits ⏲ Last 1 month timeshift -2M/M



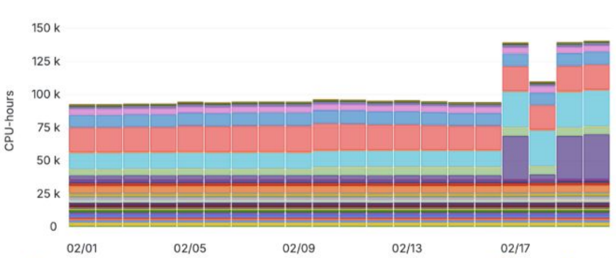

| Name | Mean | Max |
|---|---|---|
| vcluster--usdf-prompt-processing | 9.56 k | 34.7 k |
| vcluster--usdf-rapid-analysis | 23.9 k | 27.4 k |
| vcluster--usdf-rsp | 19.7 k | 23.0 k |
| vcluster--usdf-rsp-dev | 8.58 k | 9.11 k |

Container cpu limits ⏲ Last 1 month timeshift -1M/M




| Name | Mean | Max |
|---|---|---|
| vcluster--usdf-prompt-processing | 19.9 k | 58.4 k |
| vcluster--usdf-rapid-analysis | 21.1 k | 22.5 k |
| vcluster--usdf-rsp | 17.9 k | 20.6 k |
| vcluster--usdf-rsp-dev | 9.32 k | 10.5 k |

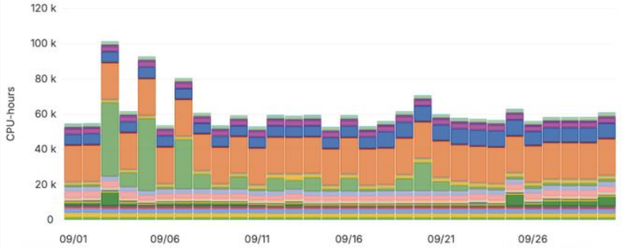
Container cpu limits ⏲ Last 1 month timeshift -0M/M




| Name | Mean | Max |
|---|---|---|
| vcluster--usdf-prompt-processing | 8.25 k | 34.3 k |
| vcluster--usdf-rapid-analysis | 15.6 k | 27.4 k |
| vcluster--usdf-rsp | 19.1 k | 20.2 k |
| vcluster--usdf-rsp-dev | 9.75 k | 10.2 k |

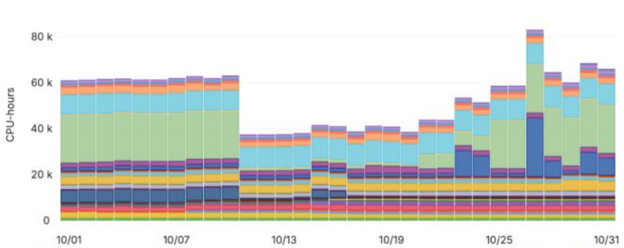
# Kubernetes CPU-Hour usage from kube_pod_requests* (last 6 months)

**Container cpu request** Last 1 month timeshift -5M/M



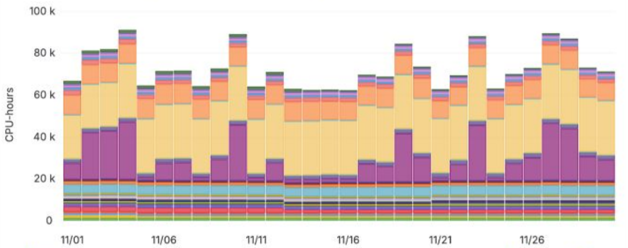

| Name | Mean | Max |
|---|---|---|
| vcluster--usdf-prompt-processing | 7.32 k | 42.0 k |
| vcluster--usdf-rapid-analysis | 20.9 k | 21.0 k |
| vcluster--usdf-rsp | 7.31 k | 9.19 k |
| vcluster--usdf-consdb | 2.17 k | 8.14 k |
| vcluster--usdf-panda | 3.18 k | 3.82 k |

**Container cpu request** Last 1 month timeshift -4M/M



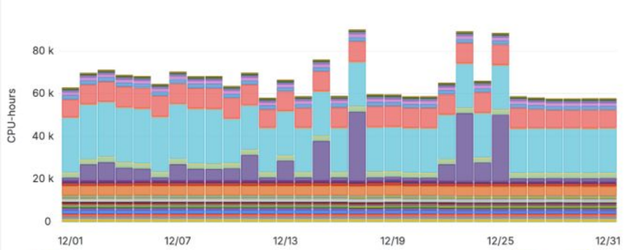

| Name | Mean | Max |
|---|---|---|
| vcluster--usdf-prompt-processing | 3.61 k | 25.8 k |
| vcluster--usdf-rapid-analysis | 12.6 k | 20.9 k |
| vcluster--usdf-rsp | 9.11 k | 9.59 k |
| vcluster--usdf-consdb | 2.53 k | 5.93 k |
| vcluster--usdf-panda | 3.53 k | 4.76 k |

**Container cpu request** Last 1 month timeshift -3M/M



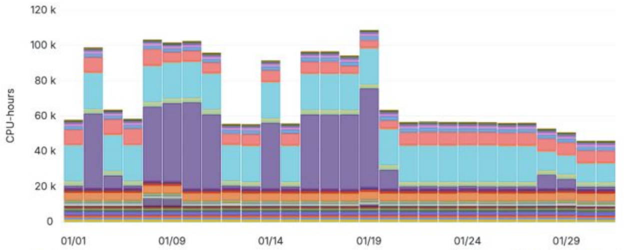

| Name | Mean | Max |
|---|---|---|
| vcluster--usdf-prompt-processing | 11.2 k | 27.4 k |
| vcluster--usdf-rapid-analysis | 25.3 k | 25.9 k |
| vcluster--usdf-rsp | 8.97 k | 9.47 k |
| vcluster--usdf-panda | 4.41 k | 4.56 k |
| vcluster--usdf-butler-dev | 2.12 k | 2.67 k |

**Container cpu request** Last 1 month timeshift -2M/M



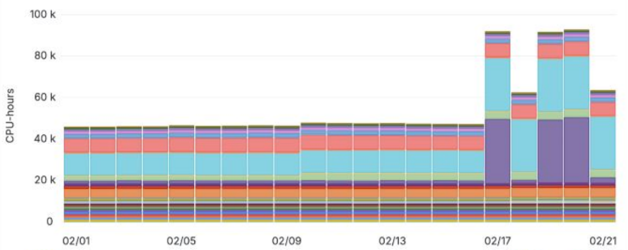

| Name | Mean | Max |
|---|---|---|
| vcluster--usdf-prompt-processing | 7.47 k | 32.6 k |
| vcluster--usdf-rapid-analysis | 22.4 k | 25.9 k |
| vcluster--usdf-rsp | 9.18 k | 9.72 k |
| vcluster--usdf-panda | 4.56 k | 4.57 k |
| vcluster--usdf-prompt-processing-dev | 2.35 k | 2.43 k |

**Container cpu request** Last 1 month timeshift -1M/M



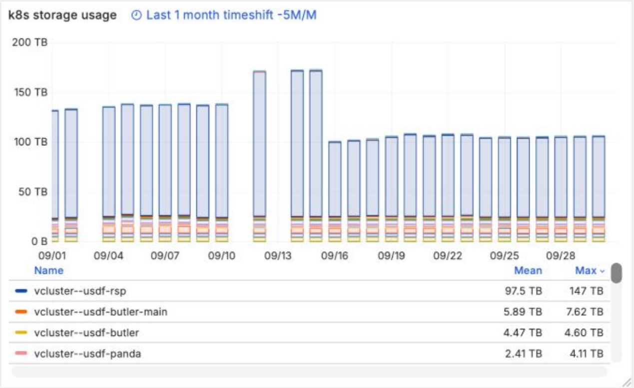

| Name | Mean | Max |
|---|---|---|
| vcluster--usdf-prompt-processing | 17.8 k | 56.3 k |
| vcluster--usdf-rapid-analysis | 19.6 k | 21.0 k |
| vcluster--usdf-rsp | 6.76 k | 9.11 k |
| vcluster--usdf-consdb | 991 | 5.02 k |
| vcluster--usdf-panda | 4.57 k | 4.57 k |

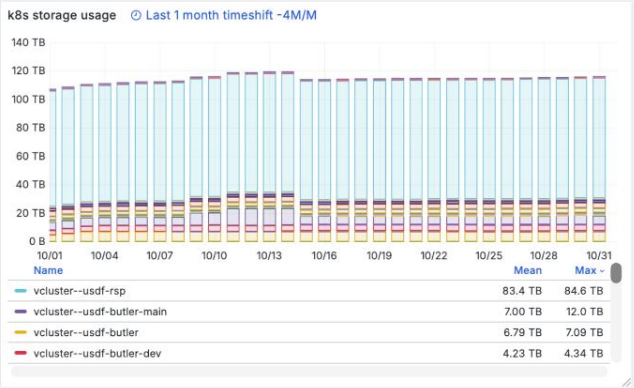
**Container cpu request** Last 1 month timeshift -0M/M

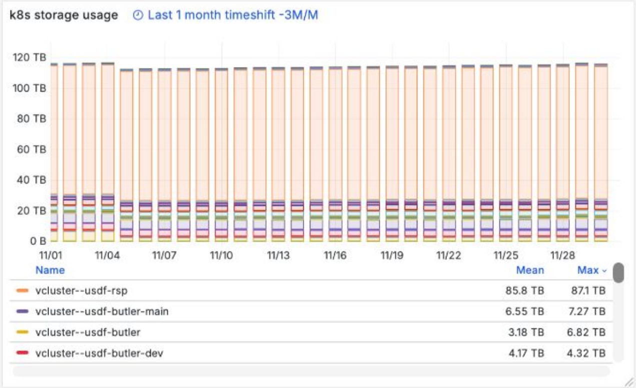


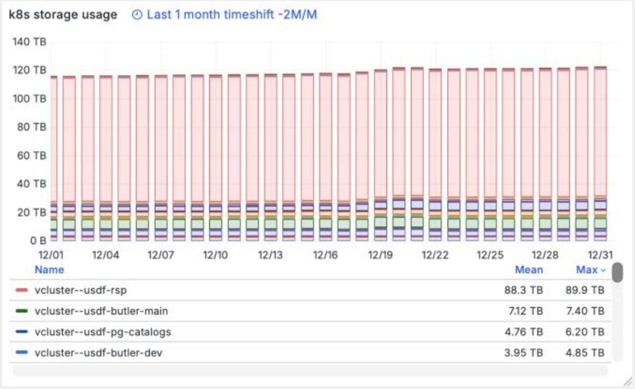

| Name | Mean | Max |
|---|---|---|
| vcluster--usdf-prompt-processing | 5.97 k | 31.9 k |
| vcluster--usdf-rapid-analysis | 14.7 k | 25.9 k |
| vcluster--usdf-rsp | 6.86 k | 7.14 k |
| vcluster--usdf-panda | 4.57 k | 4.57 k |
| vcluster--usdf-prompt-processing-dev | 3.10 k | 3.70 k |

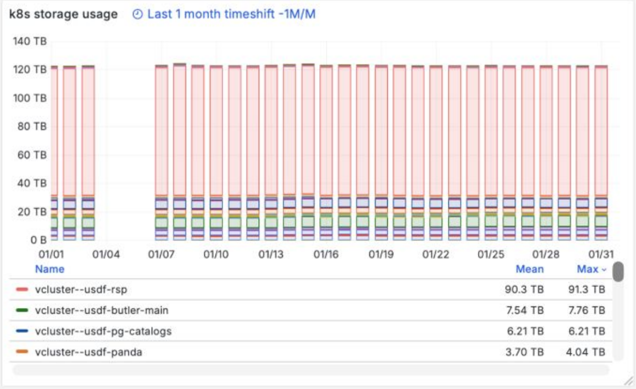
# Kubernetes storage usage (last 6 months)

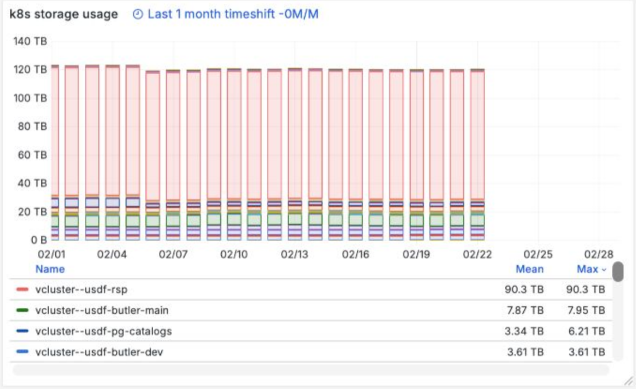


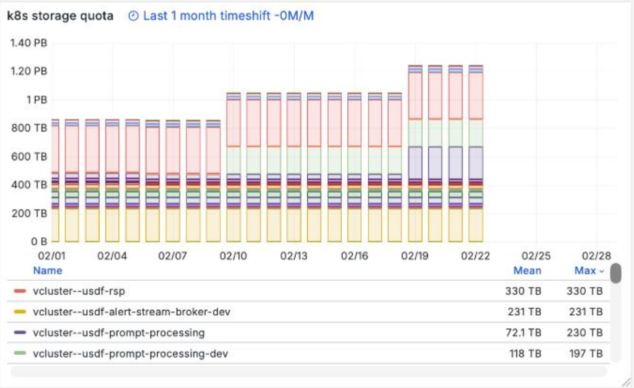



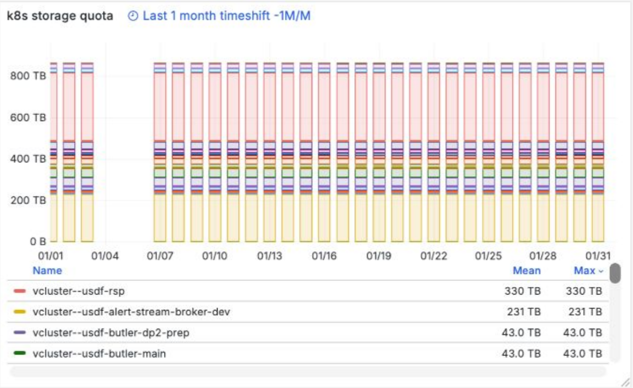



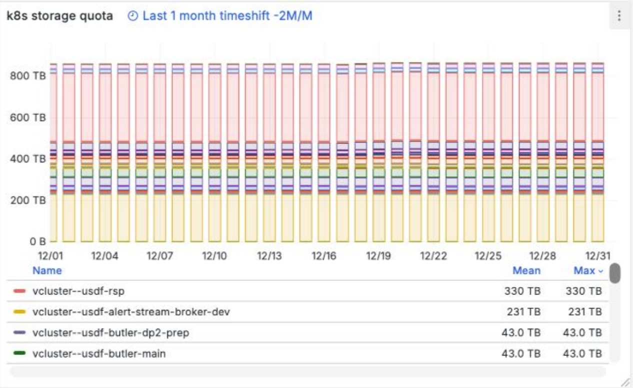



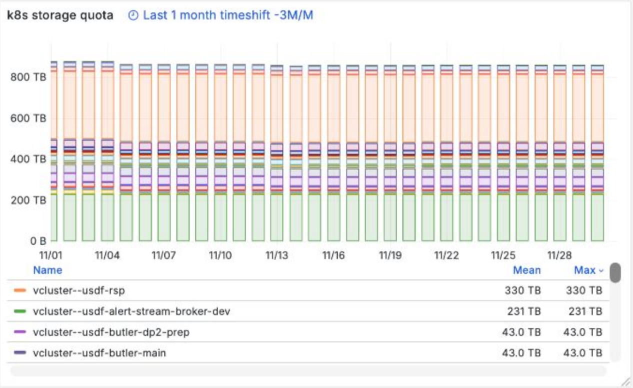



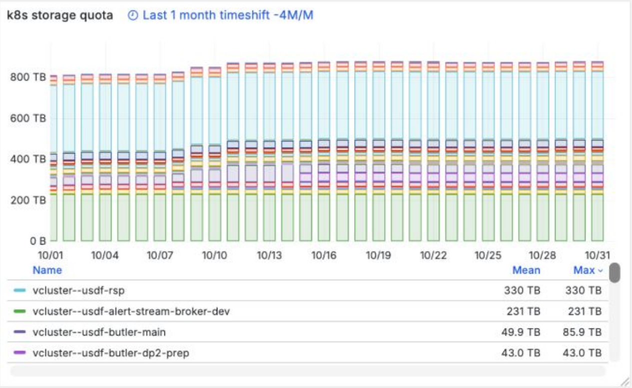
# Kubernetes storage quota usage (last 6 months)

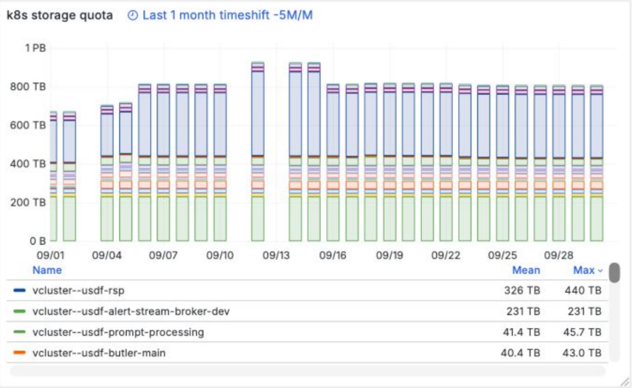


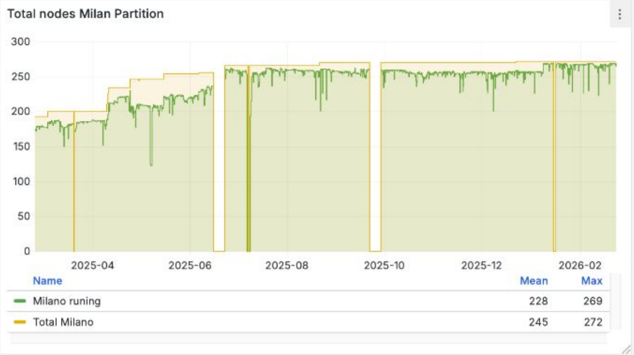



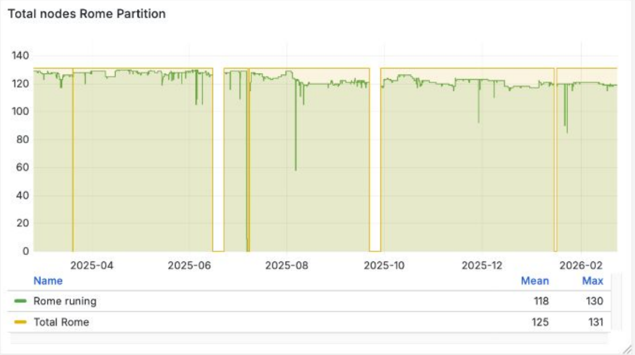



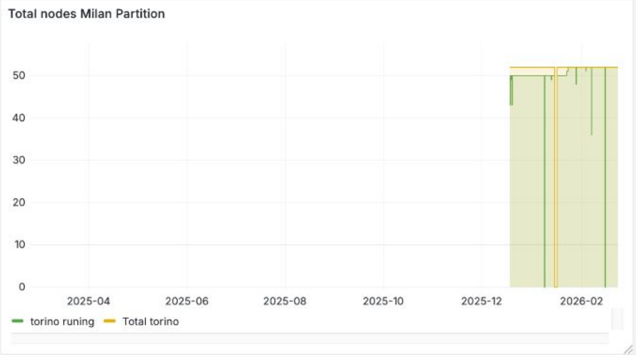



# Batch nodes - Total available nodes (1y)



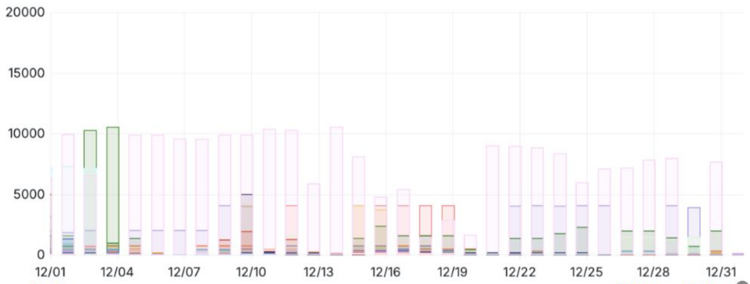

# Batch node description

- Rome: 130 Nodes. 120Cores/480GB
- Milan: 272 Nodes, 120Cores/480GB
  - 4 Nodes 120C/1920G* (sdfmilan[269-272])
- Torino: 52 Nodes. 120Cores/720GB

# Batch Max,avg cpu core per rubin account (2 months)

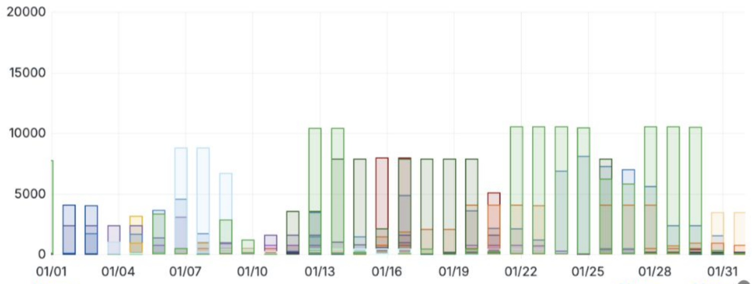
Max

max cpu core usage ⏲ timeshift -1M/M




| Name | Mean | Max |
|---|---|---|
| maxrubin:default milano preemptable leget | 6020 | 10560 |
| maxrubin:production milano normal lsstsvc1 | 6956 | 10560 |
| maxrubin:developers milano normal yusra | 2771 | 7320 |
| maxrubin:commissioning milano normal lskelvin | 1218 | 5010 |

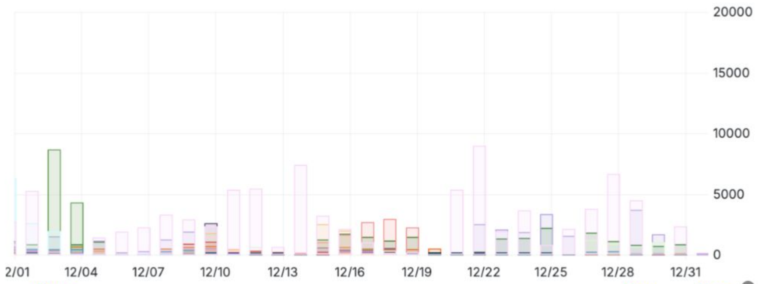
max cpu core usage ⏲ timeshift -0M/M




| Name | Mean | Max |
|---|---|---|
| maxrubin:production milano normal lsstsvc1 | 4301 | 10560 |
| maxrubin:developers milano normal yusra | 4253 | 8800 |
| maxrubin:commissioning milano normal jkurla | 3311 | 8128 |
| maxrubin:commissioning milano normal lskelvin | 4266 | 8010 |

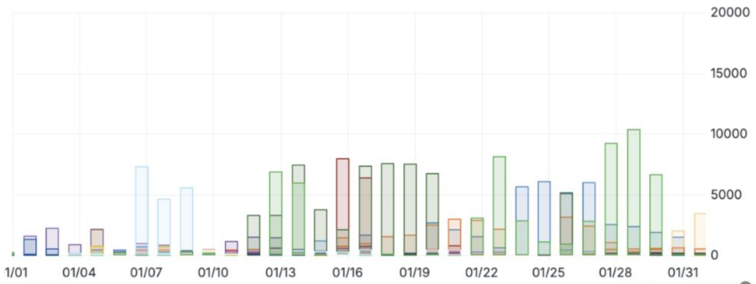
Avg

avg cpu core usage ⏲ timeshift -1M/M



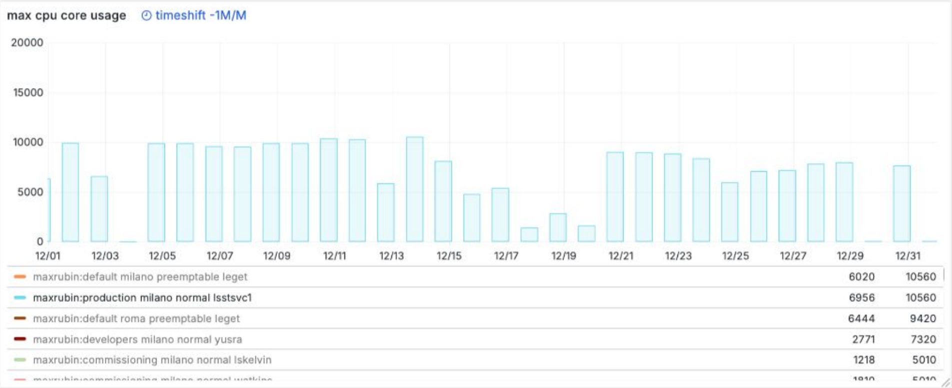

| Name | Mean | Max |
|---|---|---|
| mean rubin:production milano normal lsstsvc1 (right y-axis) | 2780 | 8977 |
| mean rubin:default milano preemptable leget (right y-axis) | 3758 | 8704 |
| mean rubin:developers milano normal yusra (right y-axis) | 1423 | 6341 |
| mean rubin:default milano preemptable jneveu (right y-axis) | 1577 | 3694 |

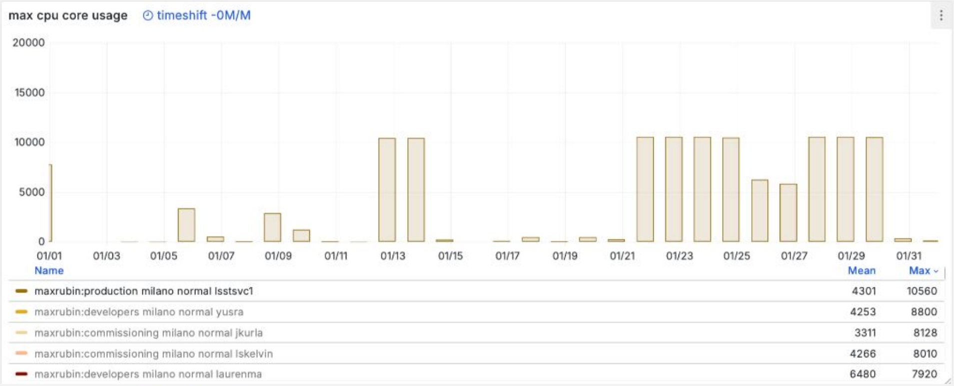
avg cpu core usage ⏲ timeshift -0M/M

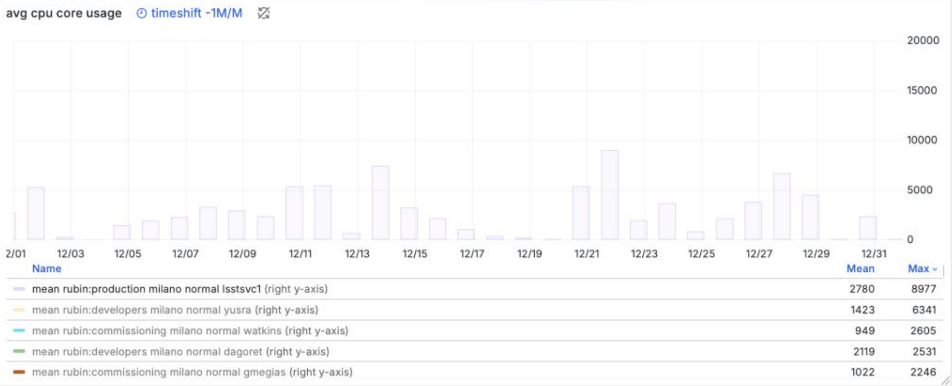


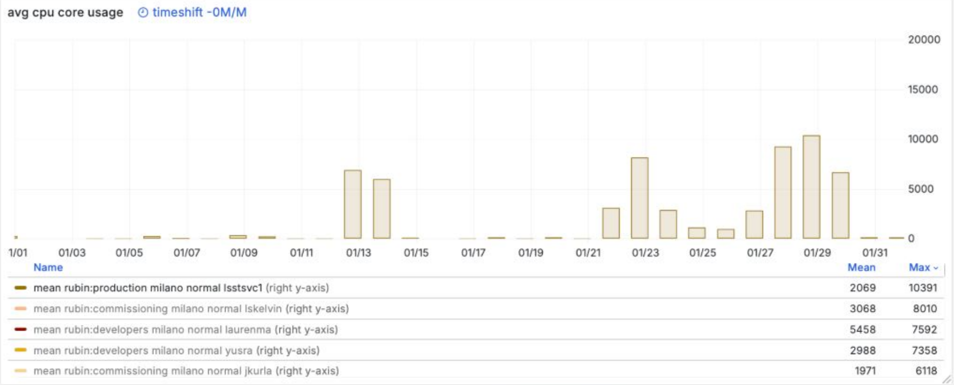

| Name | Mean | Max |
|---|---|---|
| mean rubin:production milano normal lsstsvc1 (right y-axis) | 2069 | 10391 |
| mean rubin:commissioning milano normal lskelvin (right y-axis) | 3068 | 8010 |
| mean rubin:developers milano normal laurenma (right y-axis) | 5458 | 7592 |
| mean rubin:developers milano normal yusra (right y-axis) | 2988 | 7358 |

# Batch Max,avg cpu core for rubin:pruduction (2 months)



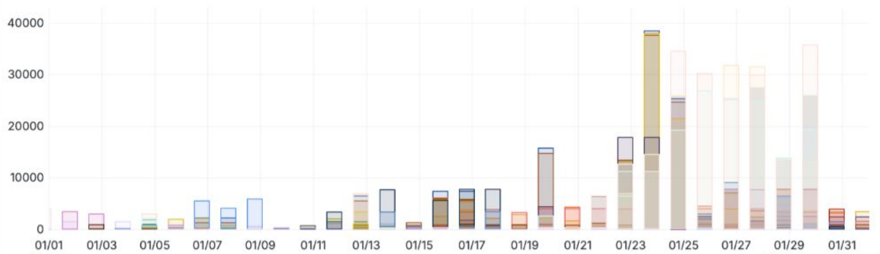



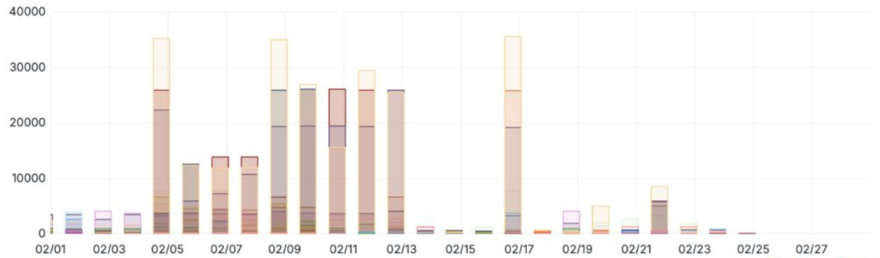
# Pending batch resources

max cpu core not running — Last 1 month timeshift -1M/M



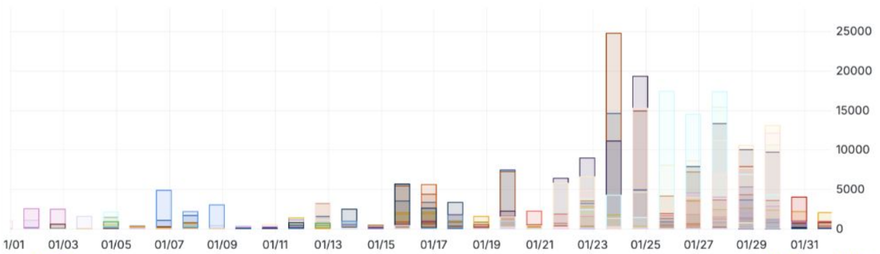

| Name | Mean | Max |
|---|---|---|
| maxrubin:commissioning milano normal jkurla AssocGrpNodeLimit | 8902 | 38592 |
| maxrubin:commissioning milano normal jkurla Priority | 6192 | 38112 |
| maxrubin:commissioning milano normal jkurla None | 7116 | 37744 |
| maxrubin:production milano normal lsstsvc1 Priority | 7118 | 35860 |
| maxrubin:production milano normal lsstsvc1 AssocGrpNodeLimit | 10728 | 31890 |

max cpu core not running — Last 1 month timeshift -0M/M



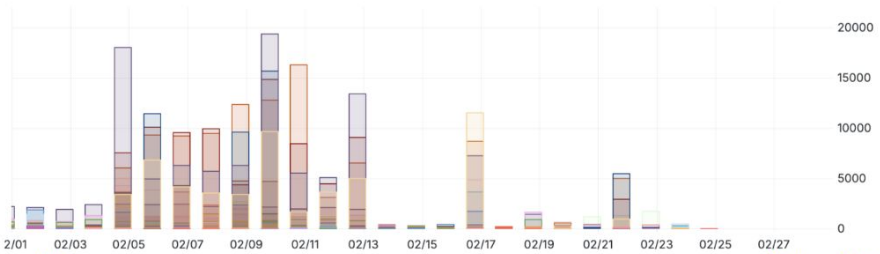

| Name | Mean | Max |
|---|---|---|
| maxrubin:production milano normal lsstsvc1 Priority | 10250 | 35640 |
| maxrubin:production milano normal lsstsvc1 AssocGrpNodeLimit | 11809 | 26109 |
| maxrubin:production milano normal lsstsvc1 AssocGrpCpuLimit | 18890 | 26070 |
| maxrubin:production milano normal lsstsvc1 AssocGrpMemLimit | 18271 | 26070 |
| maxrubin:production milano normal lsstsvc1 None | 7029 | 25960 |

mean cpu core not running — Last 1 month timeshift -1M/M



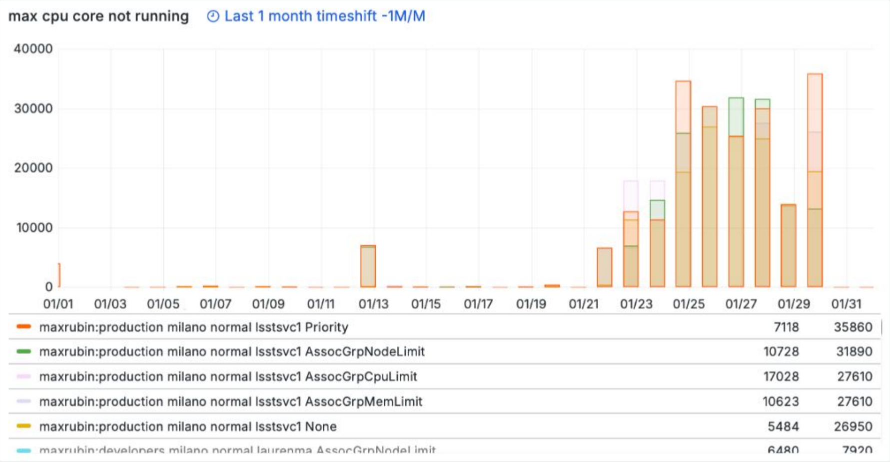

| Name | Mean | Max |
|---|---|---|
| meanrubin:commissioning milano normal jkurla None (right y-axis) | 4154 | 24796 |
| meanrubin:production milano normal lsstsvc1 AssocGrpCpuLimit (right y-axis) | 9947 | 19391 |
| meanrubin:production milano normal lsstsvc1 None (right y-axis) | 2485 | 17506 |
| meanrubin:production milano normal lsstsvc1 AssocGrpMemLimit (right y-axis) | 4456 | 15462 |
| meanrubin:production milano normal lsstsvc1 Priority (right y-axis) | 1947 | 15316 |

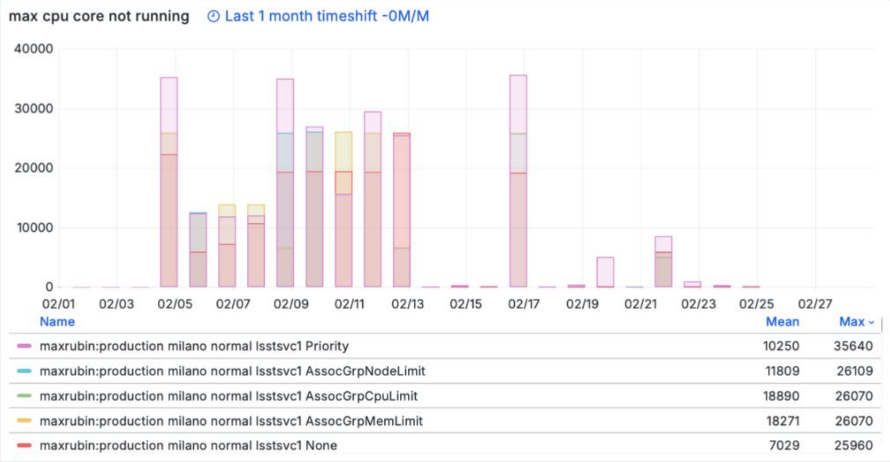
mean cpu core not running — Last 1 month timeshift -0M/M

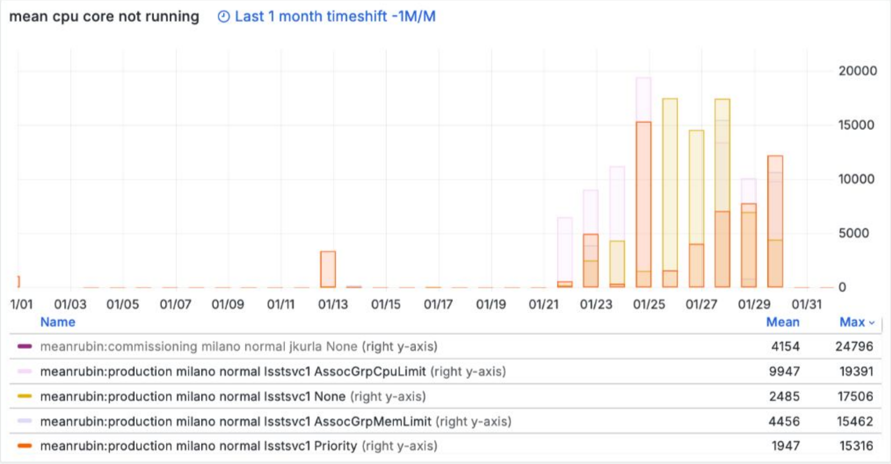


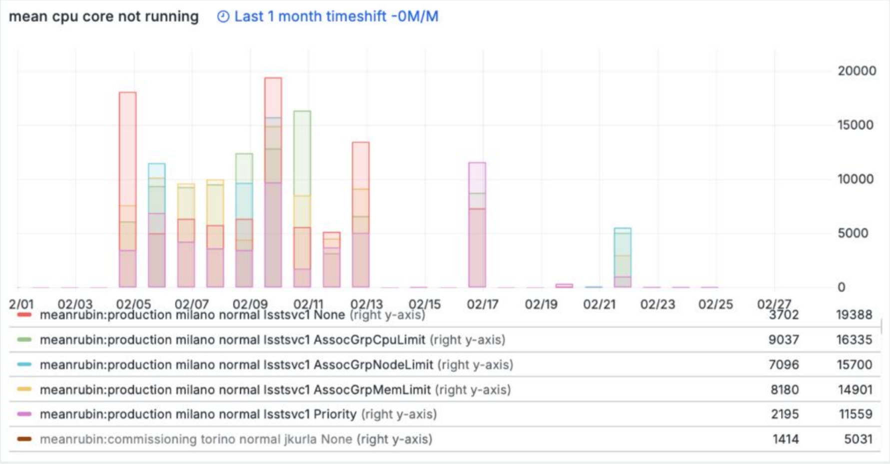

| Name | Mean | Max |
|---|---|---|
| meanrubin:production milano normal lsstsvc1 None (right y-axis) | 3702 | 19388 |
| meanrubin:production milano normal lsstsvc1 AssocGrpCpuLimit (right y-axis) | 9037 | 16335 |
| meanrubin:production milano normal lsstsvc1 AssocGrpNodeLimit (right y-axis) | 7096 | 15700 |
| meanrubin:production milano normal lsstsvc1 AssocGrpMemLimit (right y-axis) | 8180 | 14901 |
| meanrubin:production milano normal lsstsvc1 Priority (right y-axis) | 2195 | 11559 |

# Pending batch resources for rubin:production

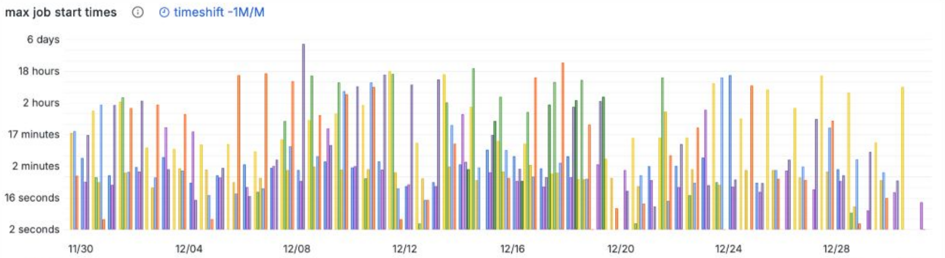


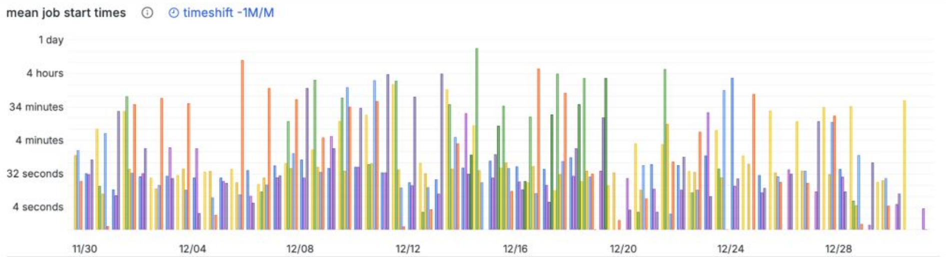



# Job queue metrics - start time - Rubin (2 months)

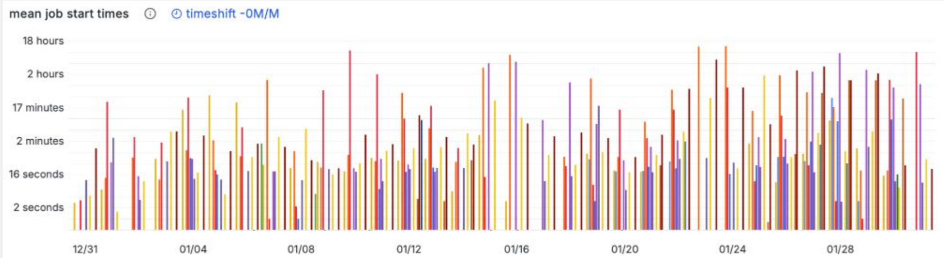



| Name | Mean | Max |
|---|---|---|
| rubin:developers milano normal | 5 hours | 4 days |
| rubin:commissioning roma normal | 3 hours | 1 day |
| rubin:commissioning ada normal | 5 hours | 21 hours |
| rubin:production milano normal | 2 hours | 18 hours |
| rubin:default milano preemptable | 35 minutes | 14 hours |


| Name | Mean | Max |
|---|---|---|
| rubin:commissioning roma normal | 4 hours | 1 day |
| rubin:commissioning milano normal | 3 hours | 1 day |
| rubin:production milano normal | 3 hours | 19 hours |
| rubin:default milano preemptable | 2 hours | 13 hours |
| rubin:developers torino normal | 2 hours | 8 hours |



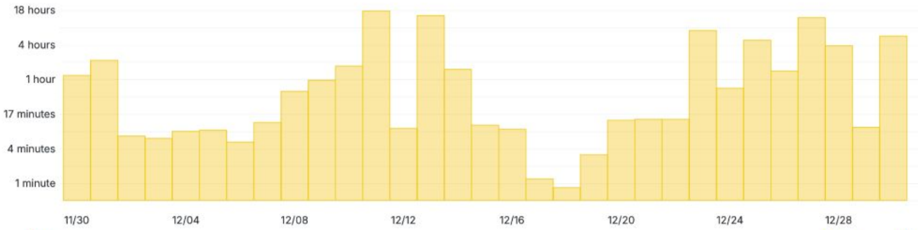

| | | |
|---|---|---|
| rubin:commissioning roma normal | 52 minutes | 10 hours |
| rubin:developers milano normal | 25 minutes | 4 hours |
| rubin:default milano preemptable | 9 minutes | 3 hours |
| rubin:developers roma normal | 46 minutes | 3 hours |
| rubin:commissioning milano normal | 14 minutes | 3 hours |
| rubin:production milano normal | 16 minutes | 2 hours |




| Name | Mean | Max |
|---|---|---|
| rubin:commissioning milano normal | 1 hour | 13 hours |
| rubin:commissioning roma normal | 56 minutes | 10 hours |
| rubin:default milano preemptable | 1 hour | 8 hours |
| rubin:production milano normal | 39 minutes | 5 hours |
| rubin:developers milano normal | 7 minutes | 2 hours |

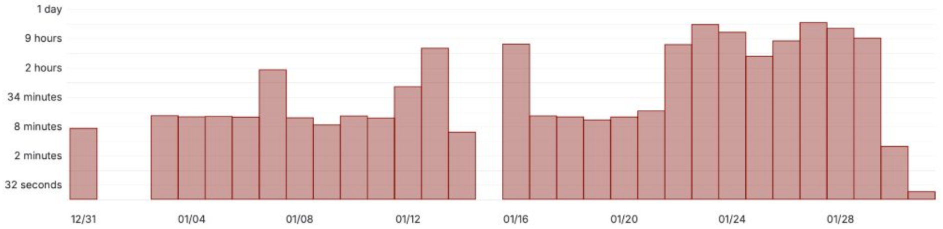
# Job queue metrics - start time - Rubin:production (2 months)

max job start times ⓘ timeshift -1M/M



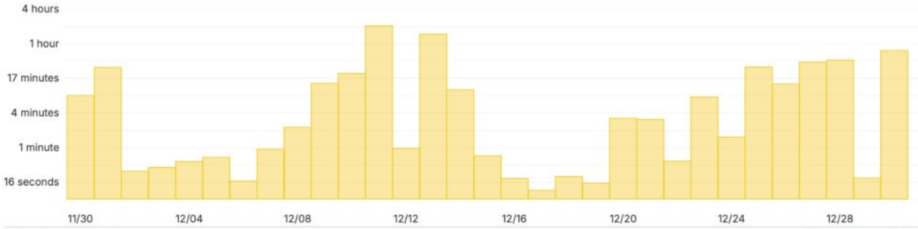

| Name | Mean | Max |
|---|---|---|
| rubin:developers milano normal | 5 hours | 4 days |
| rubin:commissioning roma normal | 3 hours | 1 day |
| rubin:commissioning ada normal | 5 hours | 21 hours |
| rubin:production milano normal | 2 hours | 18 hours |
| rubin:default milano preemptable | 35 minutes | 14 hours |

max job start times ⓘ timeshift -0M/M



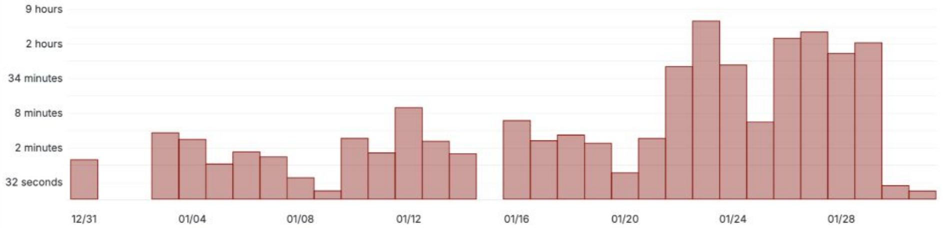

| Name | Mean | Max |
|---|---|---|
| rubin:commissioning roma normal | 4 hours | 1 day |
| rubin:commissioning milano normal | 3 hours | 1 day |
| rubin:production milano normal | 3 hours | 19 hours |
| rubin:default milano preemptable | 2 hours | 13 hours |
| rubin:developers torino normal | 2 hours | 8 hours |

mean job start times ⓘ timeshift -1M/M




| Name | Mean | Max |
|---|---|---|
| rubin:commissioning roma normal | 52 minutes | 10 hours |
| rubin:developers milano normal | 25 minutes | 4 hours |
| rubin:default milano preemptable | 9 minutes | 3 hours |
| rubin:developers roma normal | 46 minutes | 3 hours |
| rubin:commissioning milano normal | 14 minutes | 3 hours |
| rubin:production milano normal | 16 minutes | 2 hours |

mean job start times ⓘ timeshift -0M/M




| Name | Mean | Max |
|---|---|---|
| rubin:commissioning milano normal | 1 hour | 13 hours |
| rubin:commissioning roma normal | 56 minutes | 10 hours |
| rubin:default milano preemptable | 1 hour | 8 hours |
| rubin:production milano normal | 39 minutes | 5 hours |
| rubin:developers milano normal | 7 minutes | 2 hours |

# Costs

# Considerations

- All the plots are based on the dashboard https://grafana.slac.stanford.edu/d/df32i34kja96oa/review-rubin-1y?orgId=1&from=now-30d&to=now&timezone=browser&var-namespace=vcluster--usdf-butler&var-namespace=vcluster--usdf-prompt-processing&var-partition=$__all&var-qos=$__all
- If required, a reproducible script can be done doing raw queries against the prometheus/influxdb tsdb for better analysis. (but will take some time)